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Debugging and Logging Assignment

The compilation command used:

`g++ Matrix.cpp -o test -g`

Logical error:

The error found was that the output is wrong.

The output was:

33 36

66 72

The correct output is:

58 64

139 154

Debugging steps:

1. Understanding the code.
2. Calculating the expected output manually.
3. Running code and comparing with the expected output.
4. The problem is mainly in the multiply function.
5. Started gdb.
6. Putting break point on line 42 to check on the result matrix on each iteration.
7. Running program to stop at break point.
8. Then ran the program step by step.
9. On each iteration of the 3rd loop see the value of the first element that is currently being calculated.
10. Found that the value is overridden each iteration not added to the previous value.
11. Try to replace = with += in line 43.

```
adham@adham: ~/Siemens-SW-Diploma/03-Debugging_Logging/Task
File Actions Edit View Help
adham@adham: ~/Siemens-SW-.../03-Debugging_Logging/Task
Breakpoint 1 at 0x1eca: file Matrix.cpp, line 42.
(gdb) run
Starting program: /home/adham/Siemens-SW-Diploma/03-Debugging_Logging/Task/test
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Matrix 1:
1 2 3
4 5 6
Matrix 2:
7 8
9 10
11 12

Breakpoint 1, Matrix::multiply (this=0x7ffffffdc60, other=...) at Matrix.cpp:42
42     for (int k = 0; k < cols; ++k) {
(gdb) print result.data[0][0]
$1 = 0
(gdb) n
43         result.data[i][j] = data[i][k] * other.data[k][j];
(gdb) n
42     for (int k = 0; k < cols; ++k) {
(gdb) print result.data[0][0]
$2 = 7
(gdb) n
43         result.data[i][j] = data[i][k] * other.data[k][j];
(gdb) n
42     for (int k = 0; k < cols; ++k) {
(gdb) print result.data[0][0]
$3 = 18
(gdb) n
43         result.data[i][j] = data[i][k] * other.data[k][j];
(gdb) n
42     for (int k = 0; k < cols; ++k) {
(gdb) print result.data[0][0]
$4 = 33
(gdb) continue
Continuing.

Breakpoint 1, Matrix::multiply (this=0x7ffffffdc60, other=...) at Matrix.cpp:42
42     for (int k = 0; k < cols; ++k) {
(gdb) inspect result.data
$5 = std::vector of length 2, capacity 2 = {std::vector of length 2, capacity 2 = {33, 0}, std::vector of length 2, capacity 2 = {0, 0}}
(gdb) continue
Continuing.

Breakpoint 1, Matrix::multiply (this=0x7ffffffdc60, other=...) at Matrix.cpp:42
42     for (int k = 0; k < cols; ++k) {
(gdb) inspect result.data
$6 = std::vector of length 2, capacity 2 = {std::vector of length 2, capacity 2 = {33, 36}, std::vector of length 2, capacity 2 = {0, 0}}
(gdb) quit
```

GDB Commands used:

```
gdb ./test
break 42
run
n
continue
print
quit
```

Code changes made

at line 43 (multiply function loop)

instead of: `result.data[i][j] = data[i][k] * other.data[k][j]`

change it to `result.data[i][j] += data[i][k] * other.data[k][j]`

replace = by += to increment the value at each iteration instead of overriding it.